

Homework

This a growing list of problems that (hopefully) reflect the topics we've covered so far. Please clear your browser and reload this list as the semester progresses...

- (a) Please prove that, given any $\mathcal{H}$ -polyhedron $P := \{x \in \mathbb{R}^{n \times 1} \mid Ax \leq b\}$ with $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^{m \times 1}$, one can find matrices $A' \in \mathbb{R}^{(2m) \times (2n+m)}$ and $b' \in \mathbb{R}^{(2m) \times 1}$ such that $P' := \{x' \in \mathbb{R}_{\geq}^{(2n+m) \times 1} \mid A'x' = b'\}$ is non-empty if and only if P is non-empty.

(b) Following the notation of Part (a), suppose one is also given $c \in \mathbb{R}^{1 \times n}$. Can you now find $(A', b', c') \in \mathbb{R}^{(2m) \times (2n+m)} \times \mathbb{R}^{(2m) \times 1} \times \mathbb{R}^{1 \times (2n+m)}$ such that $\sup_{\substack{x \in \mathbb{R}^{n \times 1} \\ Ax \leq b}} cx = \sup_{\substack{x' \in \mathbb{R}^{(2n+m) \times 1} \\ A'x' = b'}} c'x'$?
- (a) Please prove that any polyhedron is convex.

(b) Let $\{a_1, \dots, a_t\} \subseteq \mathbb{R}^{n \times 1}$, let A be the matrix with i^{th} column a_i , and let $\hat{A}$ be the matrix with i^{th} column $\begin{bmatrix} 1 \\ a_i \end{bmatrix}$. We say $a_1, \dots, a_t$ are *affinely independent* if and only if $\hat{A}$ has right null-space $\{\mathbf{0}\}$. Please prove that $a_1, \dots, a_t$ are affinely independent $\iff t - 1$ is the dimension of the smallest affine subspace containing $\{a_1, \dots, a_t\}$.

(c) Assume $a_1, \dots, a_{n+1} \in \mathbb{R}^n$ are affinely independent. Please prove that $\alpha \in \mathbb{R}^n$ lies in the interior of the convex hull of $\{a_1, \dots, a_t\} \iff$ there are $\lambda_1, \dots, \lambda_t > 0$ with $\lambda_1 a_1 + \dots + \lambda_t a_t = \alpha$ and $\lambda_1 + \dots + \lambda_t = 1$. (Recall that $\alpha \in \mathbb{R}^n$ lies in the interior of a set $S \subseteq \mathbb{R}^n$ if and only if there is an open ball B with $\alpha \in B \subseteq S$.)
- We call a subset $S \subseteq \mathbb{R}^n$ is called a *convex cone* if and only if for all $x, y \in S$ and $\lambda, \mu \geq 0$ we have $\lambda x + \mu y \in S$.

(a) Given any $U \subseteq \mathbb{R}^n$, let $P_d(U)$ denote the collection of all polynomials in $\mathbb{R}[x_1, \dots, x_n]$ that are nonnegative on U and of degree $\leq d$. Please prove that $P_d(U)$ is a convex cone (under polynomial addition and scalar multiplication).

(b) Given any $U \subseteq \mathbb{R}^n$, let $S_{2d}(U)$ denote the collection of all polynomials in $\mathbb{R}[x_1, \dots, x_n]$, of degree $\leq 2d$, that are sums of squares of polynomials in $\mathbb{R}[x_1, \dots, x_n]$. Please prove that S_d is a convex cone (under polynomial addition and scalar multiplication).
- Suppose $S \subseteq \mathbb{R}^n$. Our original definition of $\text{Conv}(S)$ was that $\text{Conv}(S)$ is the smallest convex set (with respect to containment) containing S . Let $L(S)$ be the set of all convex linear combinations of finitely many elements of S . Please prove that $L(S)$ is convex and that any convex T containing $L(S)$ also contains $\text{Conv}(S)$. In other words, one can equivalently define $\text{Conv}(S)$ via convex linear combinations. **Hint:** The very last assertion follows easily by induction on the number of points in the convex linear combination.
- (a) Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^{m \times 1}$. Please prove *Fredholm's Theorem of the Alternative*: The linear system $Ax = b$ has *no* solution in $\mathbb{R}^n \iff$ there is a $y \in \mathbb{R}^{1 \times m}$ such that $yA = \mathbf{0}$ and $yb \neq 0$. **Hint:** This is easy if you remember reduced row echelon form from basic linear algebra.

(b) Does Fredholm's Theorem still hold if we replace $\mathbb{R}$ above by an arbitrary field K ? Please prove your answer.
- Please prove that a *univariate* polynomial $f \in \mathbb{R}[x]$ is nonnegative on all of $\mathbb{R} \iff f$ is a sum of squares, i.e., $f = g_1^2 + \dots + g_\ell^2$ identically, for some univariate polynomials $g_1, \dots, g_\ell \in \mathbb{R}[x]$. **Hint:** It helps if you use the Fundamental Theorem of Algebra to first deduce that $f = c(x - \alpha_1) \dots (x - \alpha_s)$ for some $c, \alpha_1, \dots, \alpha_s \in \mathbb{C}$, and then use the fact that all the coefficients of $\mathbb{R}$ are real to deduce further restrictions on $c, \alpha_1, \dots, \alpha_s$.
- (a) Let $f(x) := \frac{1}{89} - x^{16} + x^{49}$. Please compute the vertices of $\text{ArchNewt}(f)$, and the slopes of all the edges of $\text{ArchNewt}(f)$.

(b) Continuing Part (a), what is the cardinality of $\{\log |\zeta| \mid f(\zeta) = 0, \zeta \in \mathbb{C} \setminus \{0\}\}$?

(c) Continuing (b), verify *computationally* that, for any complex root ζ of f , there is an edge of $\text{ArchNewt}(f)$ of slope s with $|\log |\zeta| - s| < 0.001$.

(d) Now let $g(x) = 1 + x + \dots + x^8 - x^9 + \frac{1}{5}x^{10} + \frac{1}{52}x^{11} + \dots + \frac{1}{59}x^{18}$. Please find the least ε you can such that for any complex root ζ of g , there is an edge of $\text{ArchNewt}(g)$ of slope s with $|\log |\zeta| - s| \leq \varepsilon$.

8. Suppose $a_1, \dots, a_t \in \mathbb{R}^n$ and P is the convex hull of $\{a_1, \dots, a_t\}$.
Formulate a linear program that is infeasible if and only if a_i is a vertex of P . **Hint:** Consider convex linear combinations.
9. (a) In `Matlab` say

$$\text{help linprog}$$
to learn how to solve linear programs in `Matlab`. (Alternatively, you can use any software you like, or write your own code that solves linear programs having up to dozens of variables and dozens of constraints.)
(b) Please find the vertices of the convex hull of

$$\{(0, 0, 0, 0), (16, -2, 4, -8), (1, -1, 1, -1), (1, 1, 1, 1), (16, 2, 4, 8), (81, 3, 9, 27)\}.$$
Hint: You can seek out any convex hull code you like to help, or you could just use your solutions to #8 and #9(a).
10. Please prove that (x_1, x_2, x_3) lies in the standard closed unit ball (centered at the origin) if and only if the matrix $M(x) := \begin{bmatrix} 1+x_1 & x_2 & x_3 \\ x_2 & 1-x_1 & 0 \\ x_3 & 0 & 1-x_1 \end{bmatrix}$ is positive semidefinite, i.e., $M(x)$ has all its eigenvalues nonnegative. Please also determine the rank of $M(x)$ when $x_1^2 + x_2^2 + x_3^2 = 1$.